

Stock Sentiment Analysis: Legacy News Media vs. Reddit

Vincent Sham

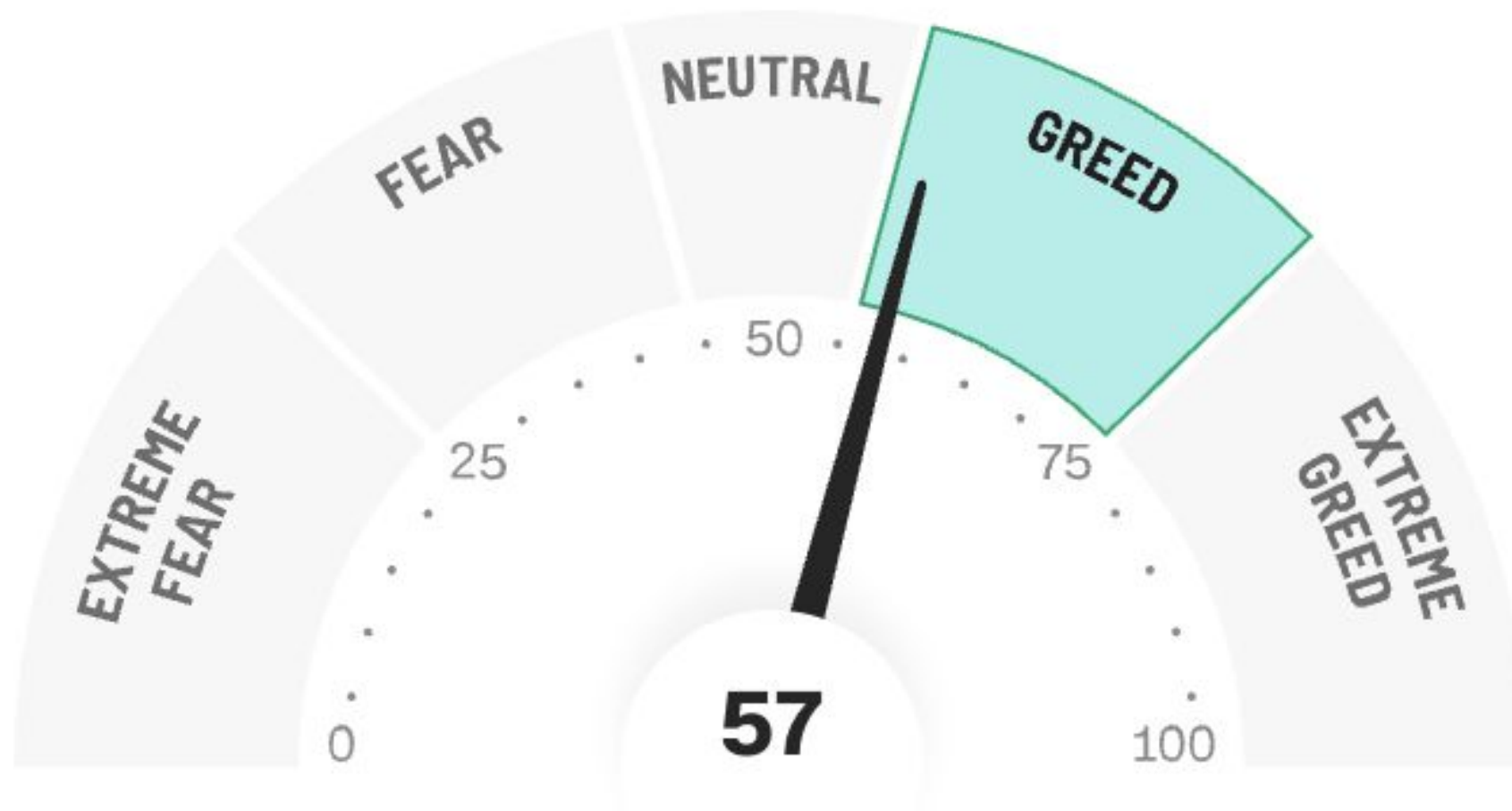
May 3, 2025



About Me

- Master of Computer Science at York University
 - Thesis: Semi-Supervised Learning with Hybrid Generative Models
- Bachelor of Science at University of Toronto
 - Actuarial Science Specialist and Statistics Major
- Pass 5 SOA exams and CFA Level 1
- 3 years of experience on Big Data Engineering

Fear and Greed Index



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Sentiment Analysis

News

vs

Reddit

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Summary

- 89% of fund managers see U.S. stocks as overvalued, the highest since 2001, as cash levels hit a 15-year low



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Plus_Refrigerator839

I'm so fucked. Till the next paycheck regards.

YOLO

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AlphaFlipper

BACK IN THE GREEN, WE ARE SO BACK.

Meme



Sentiment Analysis

News

vs

Reddit

Build Sentiment Indexes with News and Reddit

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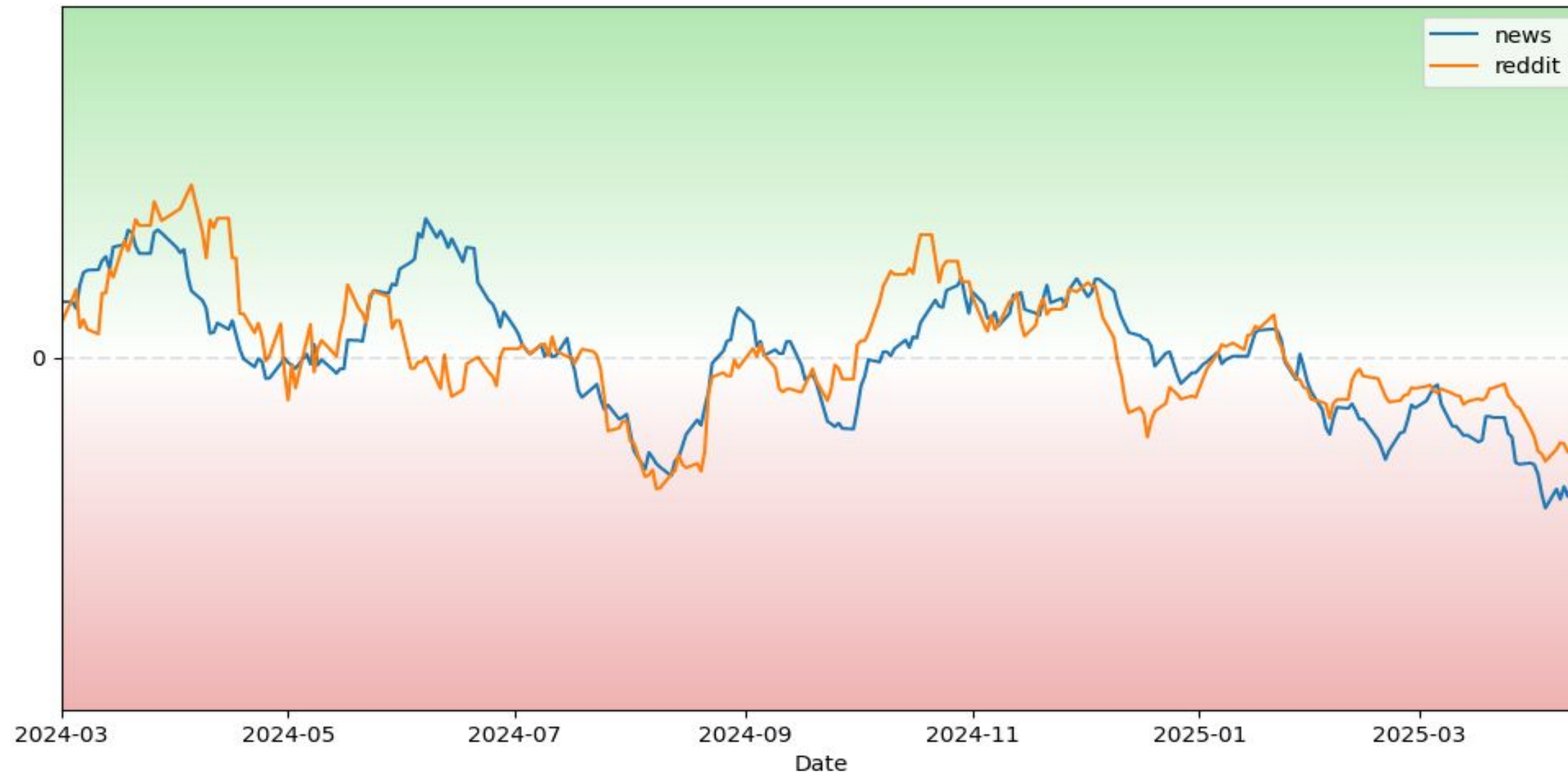
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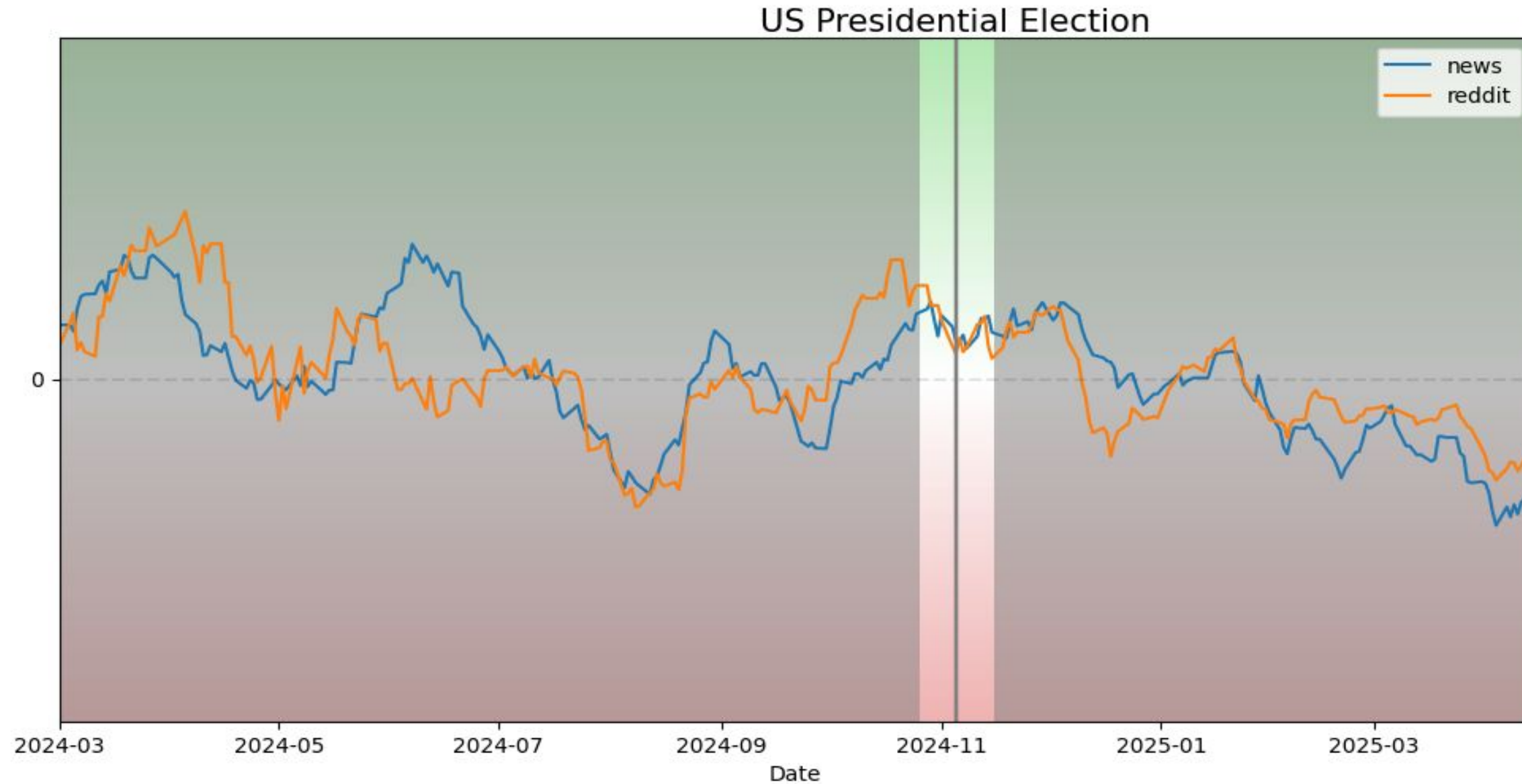
Meme



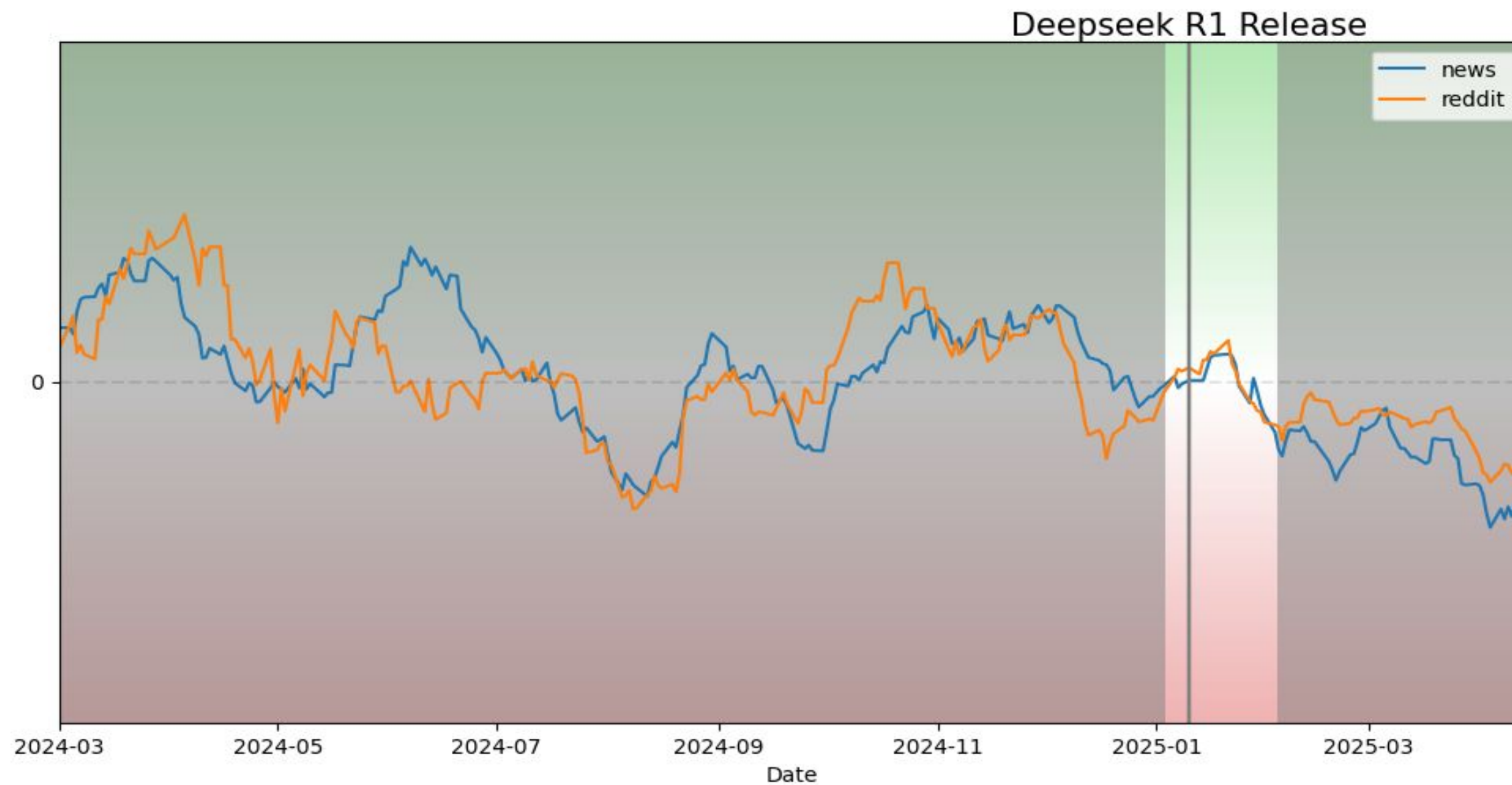
NVDA Sentiment Analysis: News vs Reddit



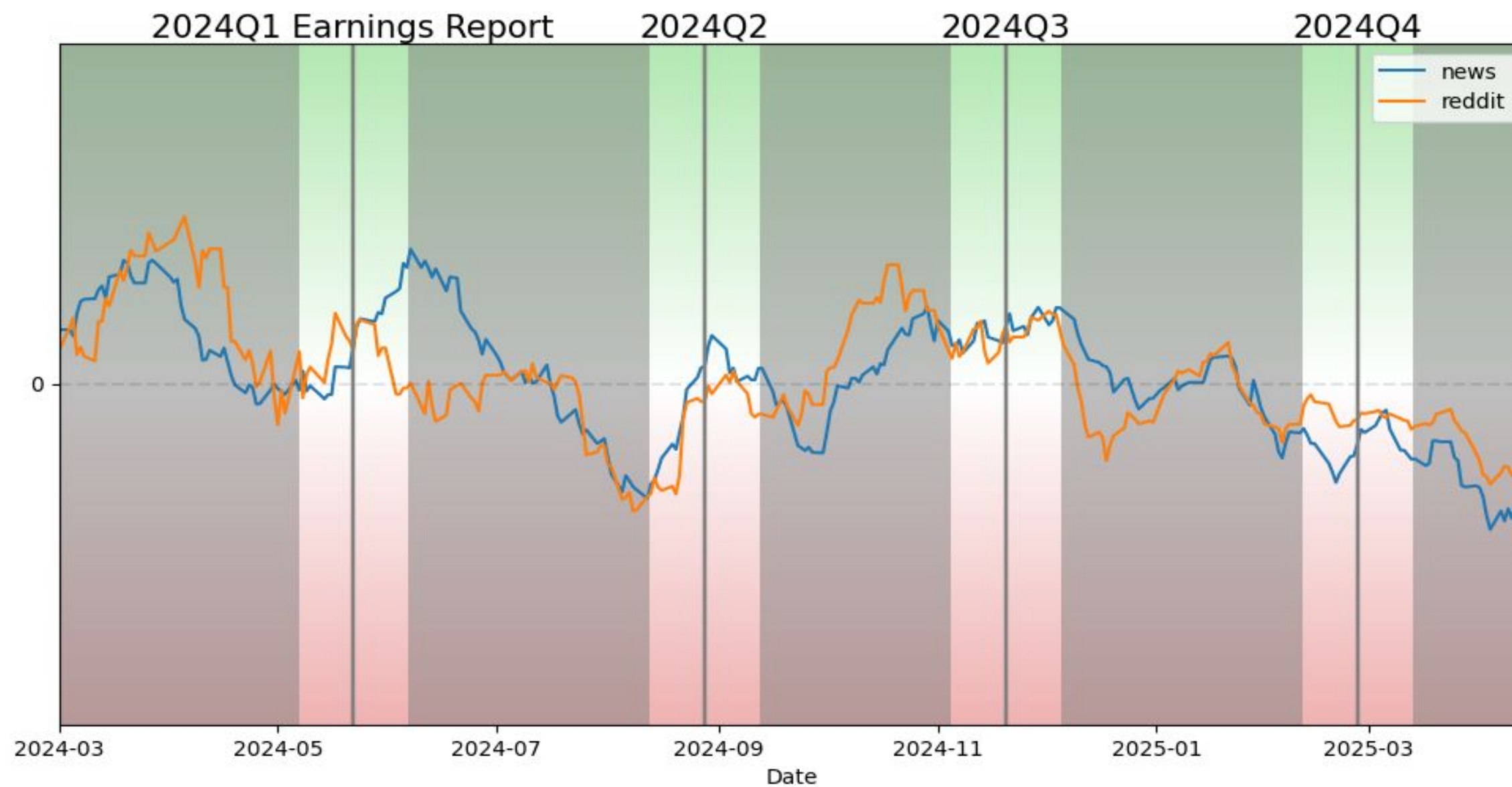
NVDA Sentiment Analysis: News vs Reddit



NVDA Sentiment Analysis: News vs Reddit



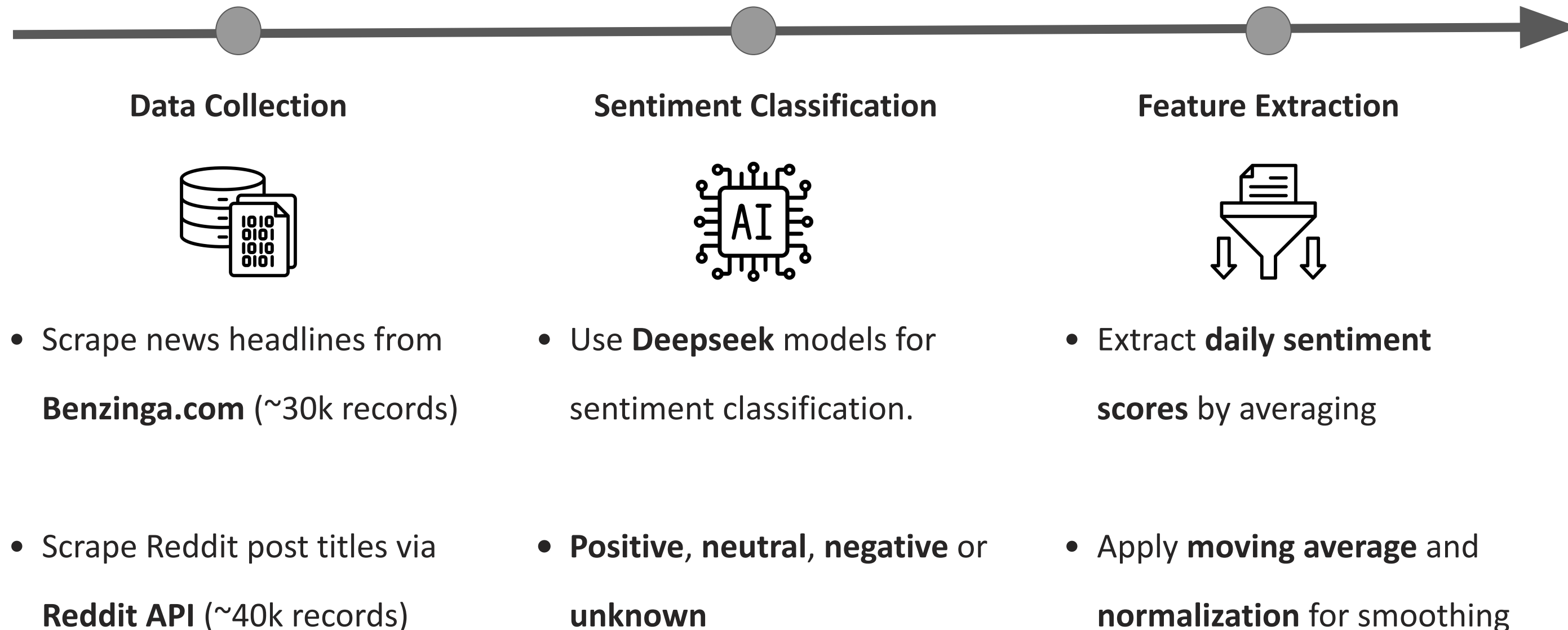
NVDA Sentiment Analysis: News vs Reddit



NVDA Sentiment Analysis: News vs Reddit



Methodology Overview



Deepseek Prompt deepseek

Prompt:

Classify the sentiment of this **Nvidia news headline** based on its potential impact on **NVDA** stock price:

positive (likely to increase)

negative (likely to decrease)

neutral (minimal impact)

or **unknown** (uncertain impact)

Headline: 'NVIDIA shares are trading lower after the com...'

Answer: negative



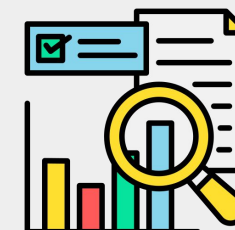
Correlation Analysis

Lead-Lag Analysis

Categorical Analysis

Modeling Experiments

Keyword Analysis



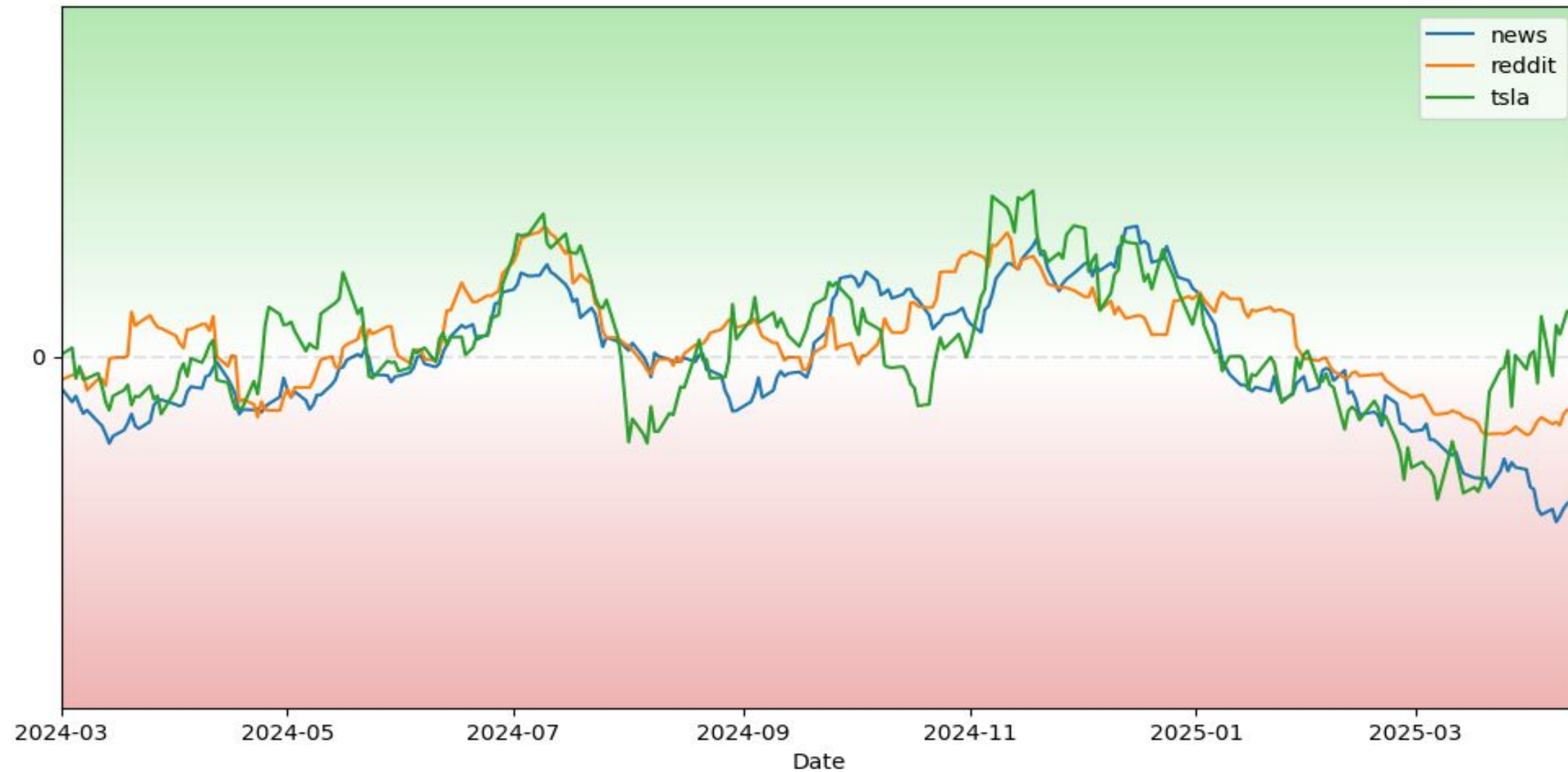


Correlation Analysis

NVDA Sentiment Analysis: News vs Reddit



TSLA Sentiment Analysis: News vs Reddit



SPY Sentiment Analysis: News vs Reddit



Cross-correlation Analysis: Sentiment vs Stock Price

	Cross-correlations					
Stocks	TSLA	AAPL	NVDA	S&P 500	PLTR	META
News vs stock price	0.69	0.74	0.62	0.45	0.04	0.08
Reddit vs stock price	0.65	0.25	0.41	0.59	0.35	0.09

Comments

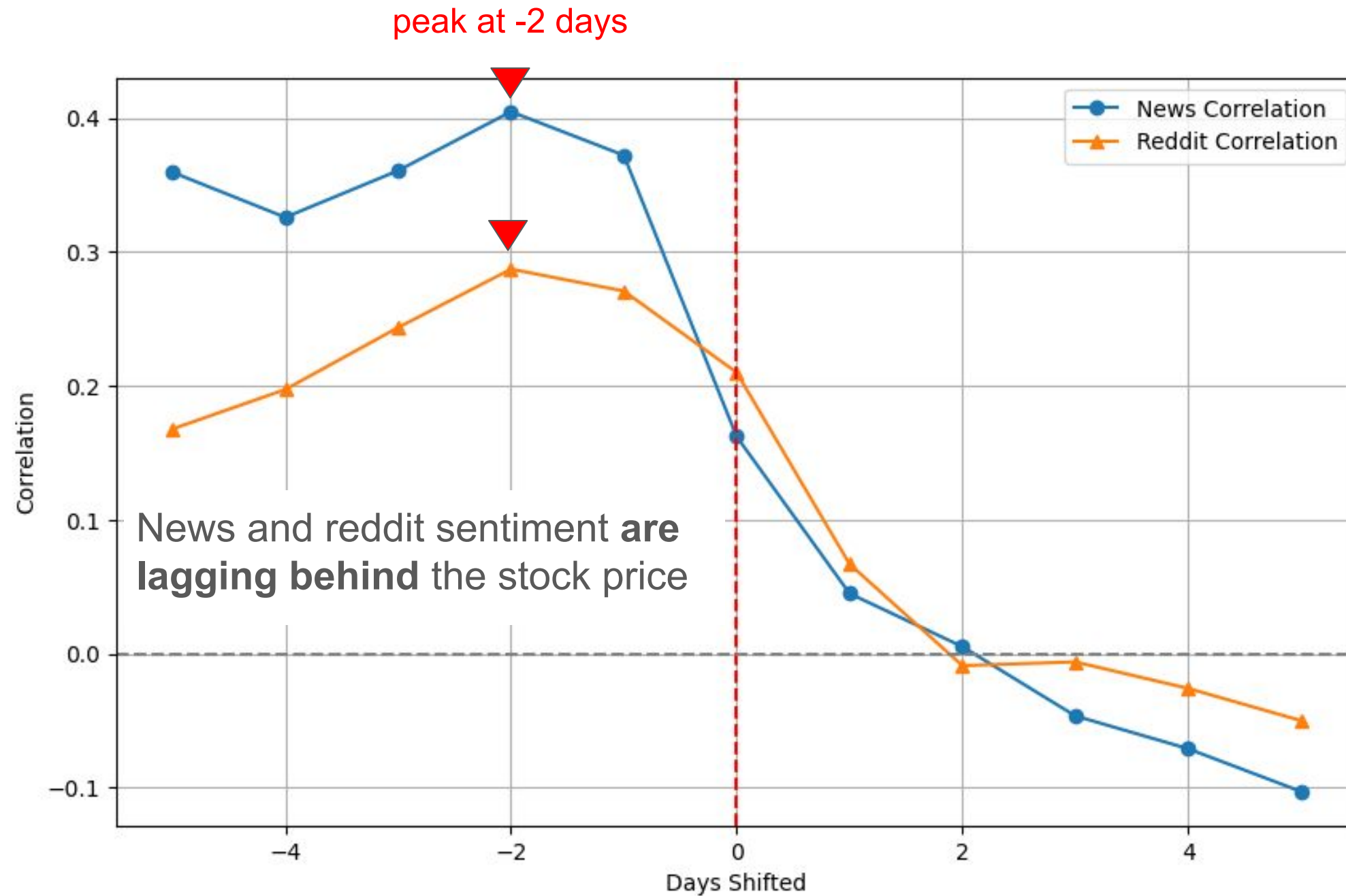
- **News**
 - News sentiment does extremely well on **Nvidia, Apple and Tesla.**
- **Reddit**
 - Reddit sentiment does better when there are **Reddit sub-communities (subreddit)** for the retail investors with lots of active users:
r/nvidia_stock, r/teslainvestorsclub, r/aapl, r/palantir_investors

Lead-Lag Analysis

Lead-Lag Analysis

- Identifying the time series X **leads or lags** the time series Y and **by how much time**

NVDA Lead-Lag Correlation Analysis



Lead-Lag Correlation Analysis

	Days Shifted					
Stocks	TSLA	AAPL	NVDA	S&P 500	PLTR	META
News vs stock price	-1	-2	-2	-1	-2	-2
Reddit vs stock price	-2	-2	-2	-1	-1	-3

- Days shifted are **negative**.
 - The sentiments are **behind** the stock prices.

Categorical Analysis

Categorical Analysis

- News and Reddit discussions on stocks are highly diverse, ranging from **factual data analysis** to **speculative opinions**.
- Categorize the news and reddit posts into four categories:
 1. data-driven
 2. fiscal-related
 3. opinion-based
 4. Other

Deepseek Prompt



Prompt:

You are an AI model that classifies the sentiment type of **financial news headlines**. Your task is to categorize each headline into one of the following sentiment types:

1. ****Data-driven**** – Headlines based on numerical data, earnings reports, market trends, or statistical insights.
2. ****Fiscal-related**** – Headlines discussing macroeconomic factors, including Federal Reserve decisions, inflation, GDP reports, interest rates, or fiscal policy changes.
3. ****Opinion-based**** – Headlines reflecting personal opinions, analyst predictions, market sentiment, or speculative analysis.
4. ****Other**** – Headlines that do not fit into the above categories.

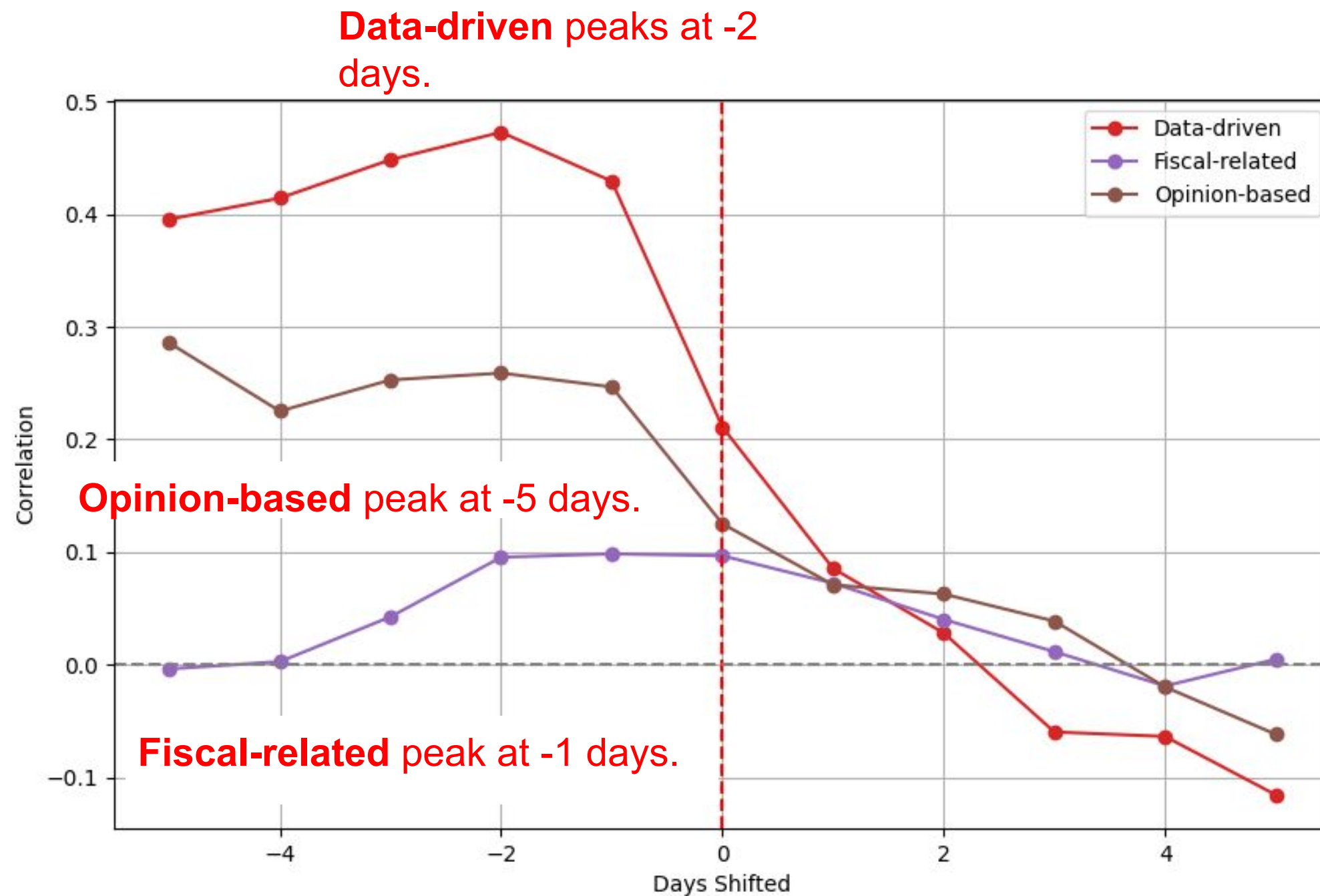
Headline: 'USA Targets DeepSeek, Weighs Restrictions On China's AI Firm That Shook Nvidia's Market Value By \$600 Billion'.

Answer: Data-driven

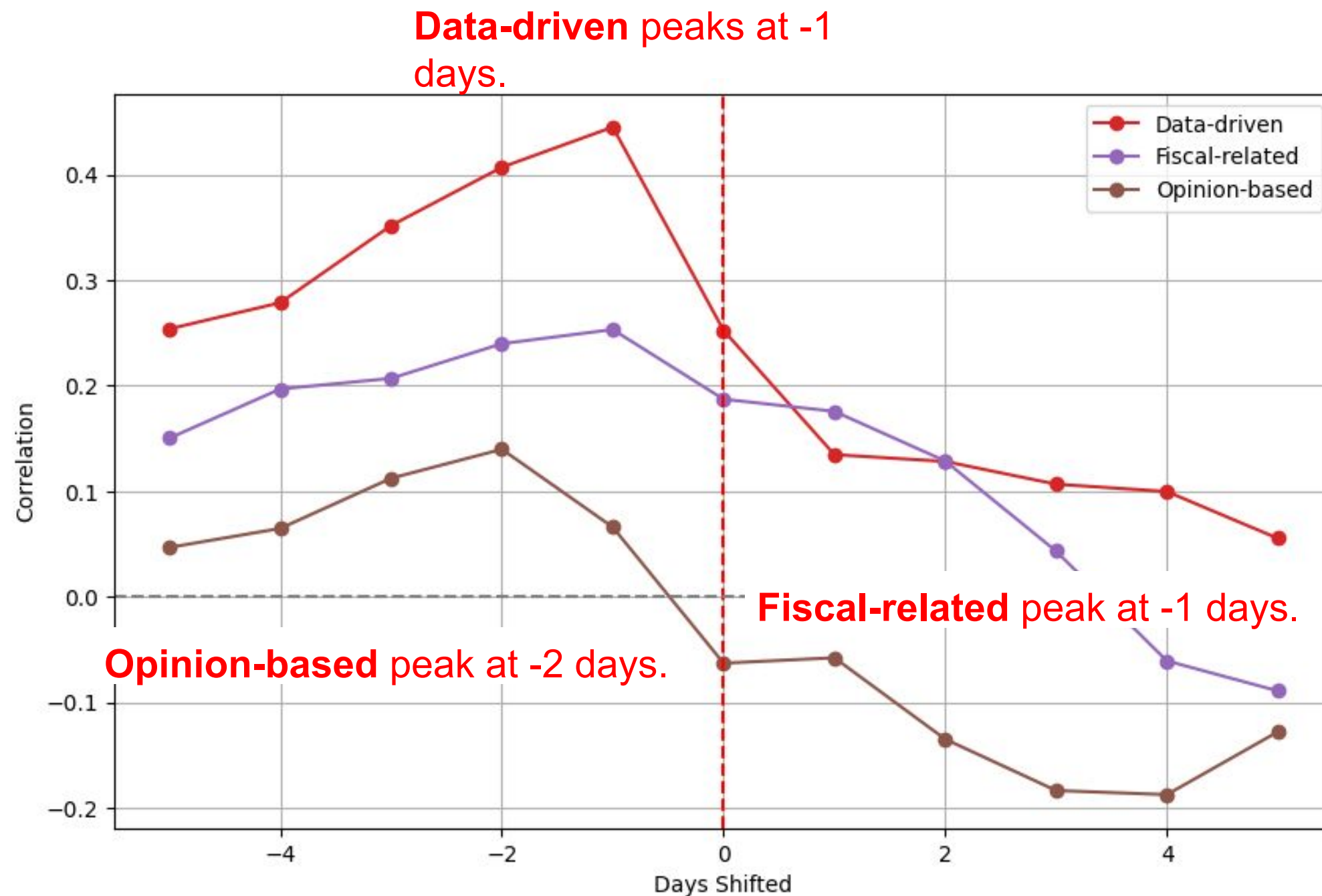
Examples

- Data-driven
 - Nvidia's AI Surge Powers 25% of S&P 500 Gains in 2024, Makes Co-founder Huang Among Fastest Growing: Report
- Fiscal-related
 - Return To 5% Bond Yields Could Tempt Investors: Are Stocks At Risk Of Losing Out?
- Opinion-based
 - I bought a stupid amount of NVDA today

NVDA Categorized News Sentiment Analysis



SPY Categorized News Sentiment Analysis



Comments

- **Data-driven** news headlines and reddit posts have higher correlations with the stock price than **fiscal-related** and **opinion-based**.
- However, the stock price **is leading** the sentiments.
 - Both the news and reddit sentiment indices **cannot predict** the stock price directly.

Modeling Experiments

Modeling Experiments

- Classify the movement of the stock price over **next five days**

$$q_t = \begin{cases} 0 & \text{if } r_t < \text{Quantile}_{0.333}(r_t) \\ 1 & \text{if } \text{Quantile}_{0.333}(r_t) \leq r_t \leq \text{Quantile}_{0.667}(r_t) \\ 2 & \text{if } r_t > \text{Quantile}_{0.667}(r_t) \end{cases}$$

where $r_t = \frac{P_{t+5} - P_t}{P_t}$

Modeling Experiments

- Input Features:
 - historical stock price movements $\mathcal{S}_{stock}$
 - historical news sentiment indexes $\mathcal{S}_{news}$
 - historical reddit sentiment indexes $\mathcal{S}_{reddit}$
- Models:
 - ARs
 - MLPs
 - Transformers

Results

	ARs	MLPs (2 layers)	MLPs (3 layers)	Transformers (1 layer)	Transformers (2 layers)	Transformers (3 layers)
S_{stock}	0.324	0.349	0.344	0.344	0.348	0.348
S_{news}	0.289	0.345	0.354	0.355	0.366	0.358
S_{reddit}	0.308	0.314	0.332	0.320	0.332	0.329

- Best performance
 - Model: Transformers
 - Input Feature: News sentiment index

Results

	ARs	MLPs (2 layers)	MLPs (3 layers)	Transformers (1 layer)	Transformers (2 layers)	Transformers (3 layers)
$S_{stock} + S_{news}$	0.301	0.349	0.360	0.369	0.362	0.366
$S_{stock} + S_{reddit}$	0.323	0.332	0.340	0.343	0.342	0.341
$S_{news} + S_{reddit}$	0.281	0.354	0.362	0.364	0.378	0.373
$S_{stock} + S_{news} + S_{reddit}$	0.319	0.362	0.361	0.360	0.366	0.365

- Best performance
 - Model: Transformers
 - Input Feature: News sentiment index + reddit sentiment index

Comments

- We hypothesize that
 - historical stock prices tend to be **backward-looking**.
 - stock prices **do not fully reflect** the information provided by sentiments.
- However, further investigation is needed to fully understand this phenomenon.

Keyword Analysis



Top Keywords (based on TF-IDF)

2024-03 - 2024-09

Before

2024-10 - 2024-12

During

2025-01 - 2025-03

After Election

News

fed says **powell** chair **biden**

News

fed says usa **trump** us

News

trump says **fed** president chair

Reddit

market spy strategy **fed** stock

Reddit

strategy market stock spy best

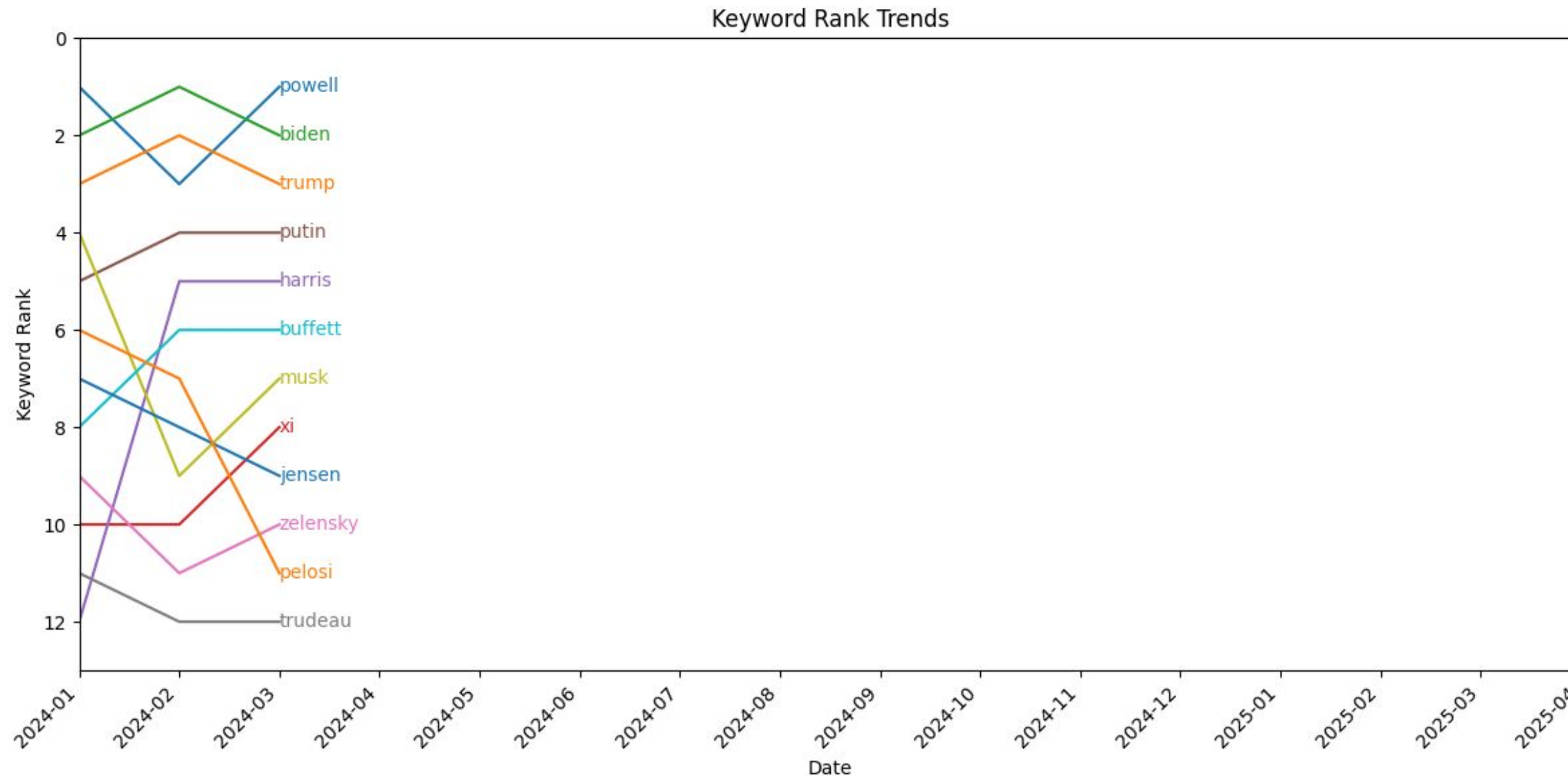
Reddit

market strategy spy **tariffs**
trump

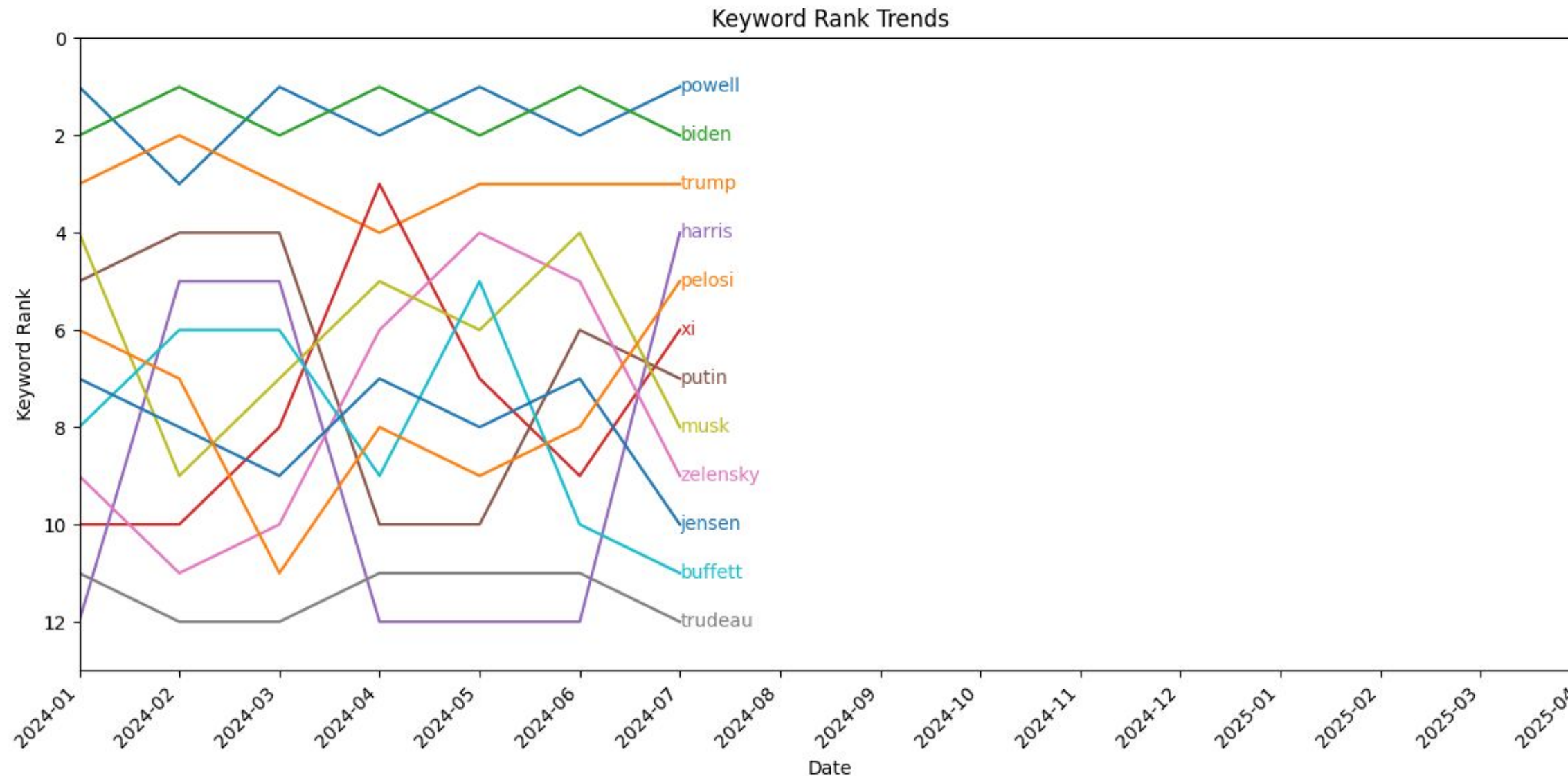
Comments

- Many top keywords are related to **governors**.
- Influential figures like **governors** and **CEOs** can greatly impact the market.

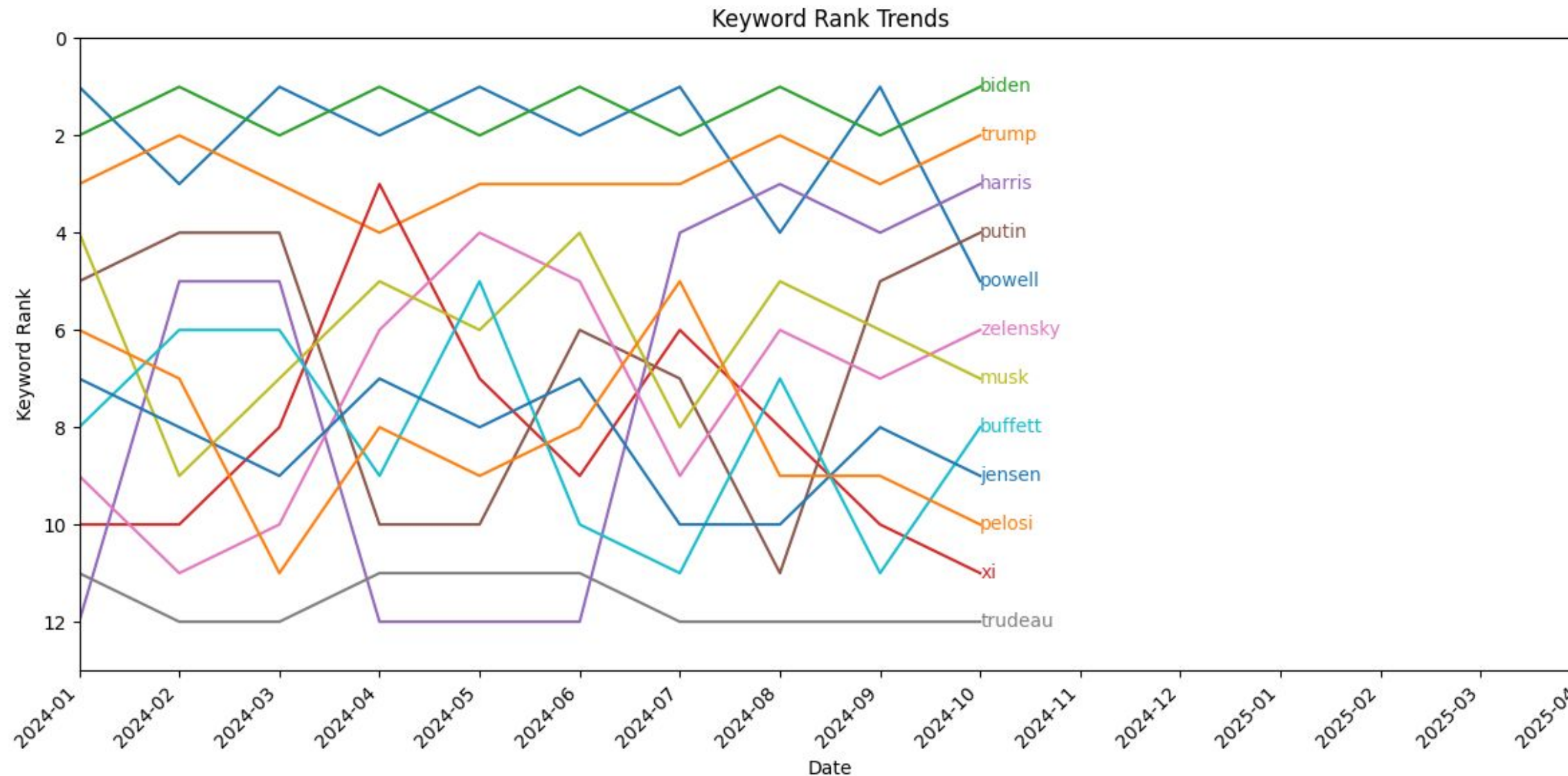
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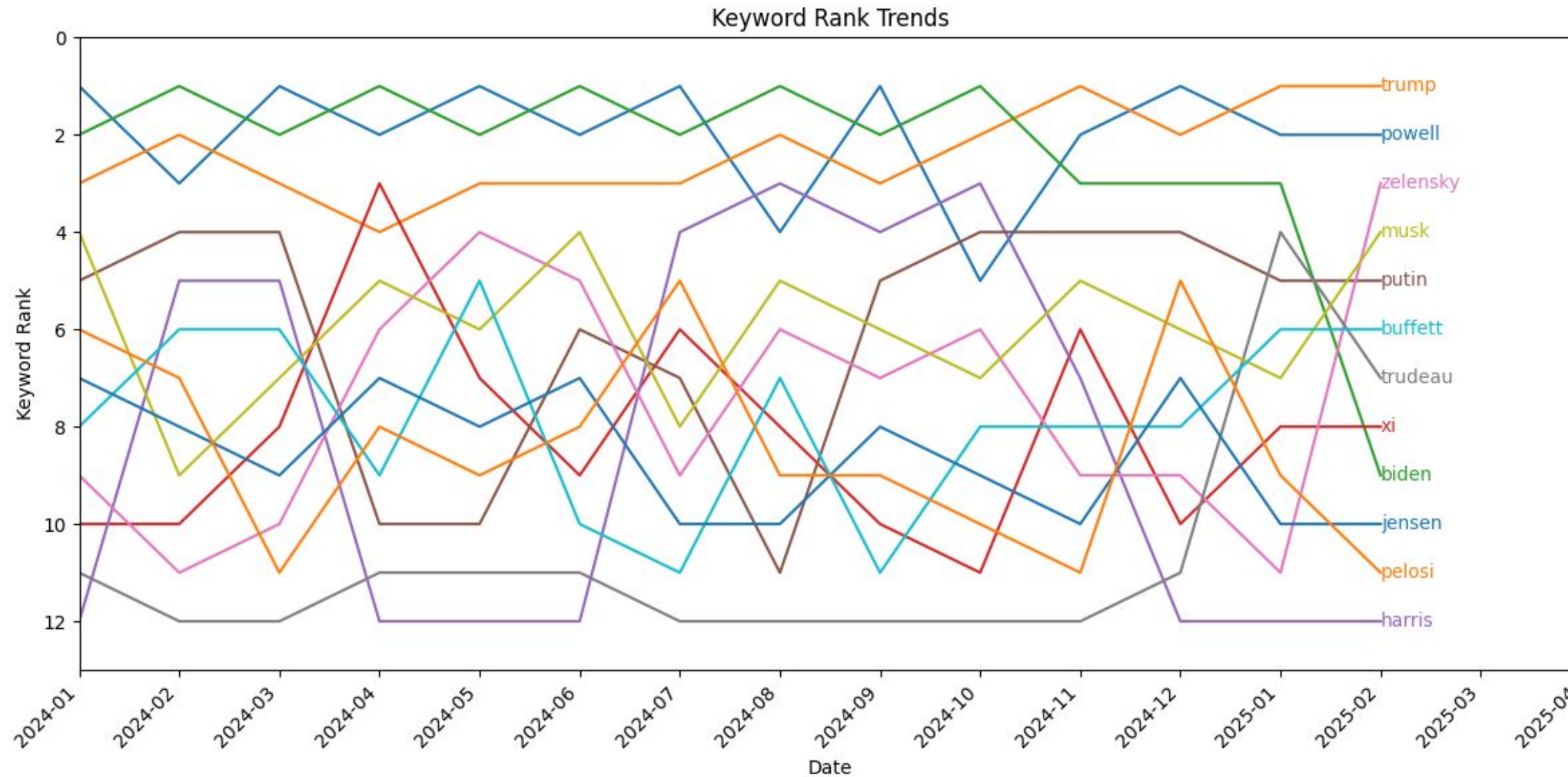
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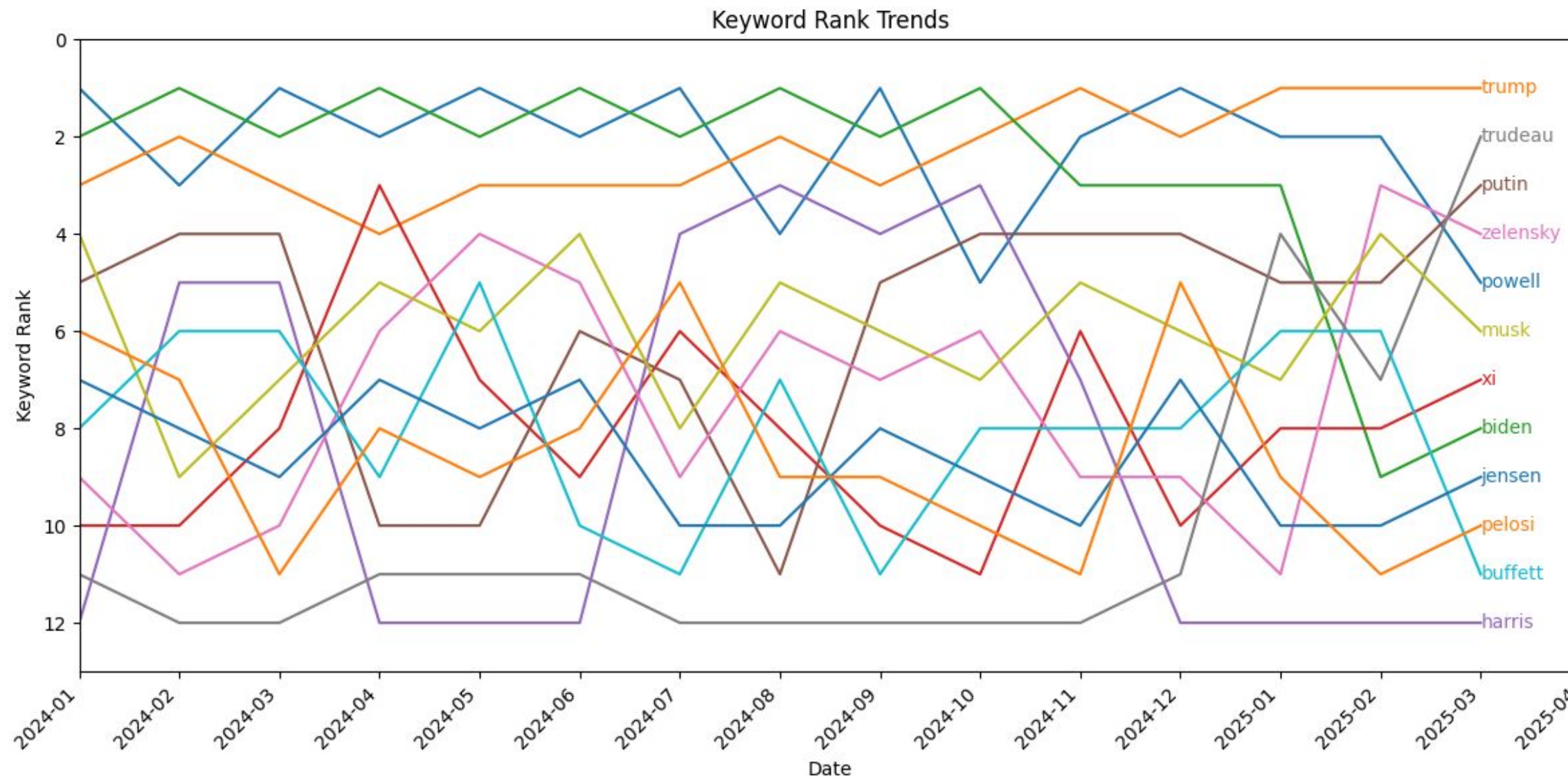
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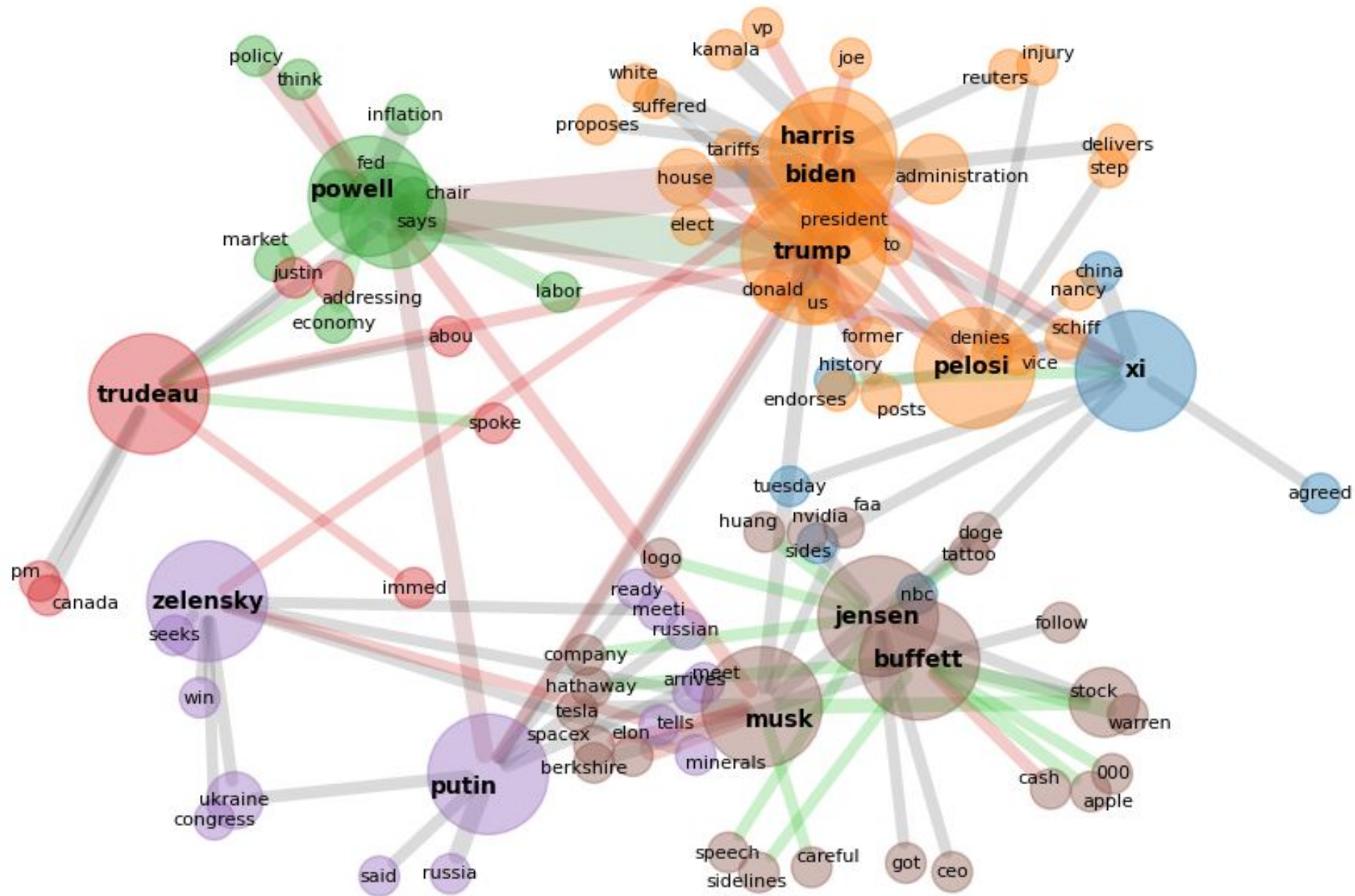
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Influential Figures in the U.S. Stock Market: Ranking Trends



Network Graph with Cluster Separation



Limitations

- **Limited Time Period (2024-2025 only)**
 - Historical news and Reddit posts are costly to access.
- **LLM Dependency**
 - Classification relies solely on DeepSeek, which may introduce biases.
- **Source Selection Bias**
 - Sentiments may differ across platforms and data sources.

Future Works

- **Expand Sentiment Analysis Methods**
 - Test **BERT**, **VADER**, and others for better accuracy.
- **Analyze Sentiment-Driven Market Reactions**
 - Identify patterns such as **FOMO-driven spikes** or **overreactions** affecting stock prices.
- **Keyword Trend Analysis**
 - Analyze how **rising** or **falling** keyword mentions relate to stock shifts.

References

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- [6] Zengzhi Wang, Qiming Xie, Yi Feng, Zixiang Ding, Zinong Yang, and Rui Xia. Is chatgpt a good sentiment analyzer? a preliminary study, 2024.

Thank you!